

- 3 Two sources of sound S_1 and S_2 emit waves of frequency 220 Hz in phase as shown in Fig. 3.1.

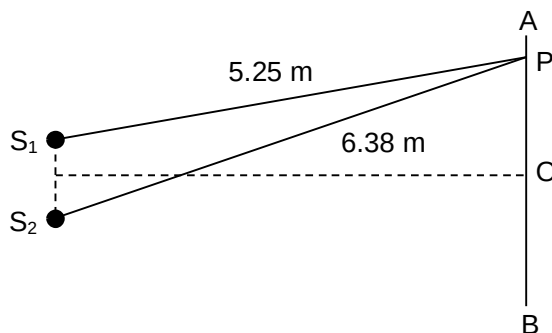


Fig. 3.1

- (a) Sound wave is a longitudinal wave.

Explain what is meant by a longitudinal wave.

.....

 [1]

- (b) At point P along the line AB , the waves are found to have a phase difference of $\frac{3}{2}\pi$ rad. S_1 and S_2 are 5.25 m and 6.38 m away from P respectively. For the sound waves emitted, determine

- (i) the wavelength,

wavelength = m [2]

- (ii) the speed.

speed = m s^{-1} [2]

- (c) Point O is equidistant from S_1 and S_2 . Sound waves from S_1 and S_2 arrive at O with intensities I and $3I$ respectively.

Determine the resultant intensity at O in terms of I .

resultant intensity = [3]

4 (a) Sound waves produced by a loudspeaker arrive at the microphone M via two different